* Percentiles

easy

1) 10, 20, 30, 40, 50

P_{25} = 15
P_{50} = 30
P_{75} = 45

medium

2) 12, 15, 18, 21, 24, 27, 30, 33, 36

P_{10}	=	~~13.5~~	12
P_{25}	=	~~19.5~~	16.5
P_{75}	=	~~28.5~~	31.5
P_{90}	=	~~34.2~~	36
P_{50}	=	~~24~~	24

3) 20, 25, 30, 35, 40, 45, 50, 55, 60, 65

P_{10}	=	20	P_{20}	=	22.5
P_{25}	=	27.5	P_{40}	=	32.5
P_{50}	=	42.5	P_{60}	=	52.5
P_{75}	=	57.5	P_{80}	=	62.5 ×
P_{90}		× 65			

Main answer

$P_{20} = \frac{20}{100} \times (10 + 1)$

$= \frac{20}{100} \times 11$

$= 0.2 \times 11$

$= \boxed{2.2}$

$25 + 0.2(5)$
$25 + 1$
$\boxed{26}$

$p40 = \frac{40}{100} \times (10+1)$

$= 0.4 \times 11$

$\boxed{4.4}$

$35 + 0.4(5)$

$= 35 + 2$

$= \boxed{37}$

$P60 = \frac{60}{100} \times (10+1)$

0.6×11

$= \boxed{6.6}$

$45 + 0.6(5)$

$45 + 3$

$= \boxed{48}$

$P80 = \frac{80}{100} \times (10+1)$

$= 0.8 \times 11$

$= \boxed{8.8}$

$55 + 0.8(5)$

$= 55 + 4$

$= \boxed{59}$

4) 100, 120, 140, 160, 180, 200, 220, 240

$p25 = 225$

$p50 = 168$

$p75 = 215$

$p25 = \frac{25}{100} \times (8+1)$

0.25×9

$= \boxed{2.25}$

$120 + 0.25(20)$

$= 120 + 5$

$= \boxed{125}$

$P50 = \frac{50}{100} \times (8+1)$

$= 0.5 \times 9$

$= \boxed{4.5}$

$160 + 0.5(20)$

$160 + 10$

$= \boxed{\cancel{000}}\ 170$

$P75 = \frac{75}{100} \times (8+1)$

$= 0.75 \times 9$

$= \boxed{6.75}$

$200 + 0.75(20)$

$200 + 15$

$= \boxed{215}$

Hard level

5] 5, 10, 15, 20, 25, 30, 35, 40, 50, 100

P10 = 5.5

P25 = ~~11.25~~ 13.75

P50 = 27.5

P75 = ~~43.75~~ ~~41.05~~ 42.5

P90 = 95

$P10 = \frac{10}{100} \times (10+1)$

$= 0.1 \times 11$

$\boxed{1.1}$

$5 + 0.1(5)$

$= 5 + 0.5$

$= \boxed{5.5}$

$P_{25} = \frac{25}{100} \times (10+1)$

$= 0.25 \times 11$

$= \boxed{2.75}$

$10 + 0.25(5)$

$10 + 2.75$

$= \boxed{13.75}$

$P_{50} = \frac{50}{100} \times (10+1)$

0.5×11

$= \boxed{5.5}$

$25 + 0.5(5)$

$= 25 + 2.5$

$= \boxed{27.5}$

$P_{75} = \frac{75}{100} \times (10+1)$

$= 0.75 \times 11$

$= \boxed{8.25}$

$40 + 0.25(10)$

$= 40 + 2.5$

$= \boxed{42.5}$

$P_{90} = \frac{90}{100} \times (10+1)$

$= 0.90 \times 11$

$= \boxed{9.9}$

$50 + 0.90(50)$

$= 50 + 45$

$= \boxed{95}$

6) 18, 22, 25, 30, 35, 40, 45, 50, 55, 60, 120

$P_{25} = \cancel{40}\ 25$
$P_{50} = \cancel{70}\ 40$
$P_{75} = \cancel{95}\ 55$
$P_{90} = 108$

$P_{25} = \frac{25}{100} \times (11+1)$

0.25×12
$= \boxed{3}$

$25 + 3(5)$
$= 25 + 15$
$\boxed{40}$

$P_{50} = \frac{50}{100} \times (11+1)$

0.50×12
$= \boxed{6}$

$40 + 6(5)$
$= 40 + 30$
$= \boxed{70}$

$P_{75} = \frac{75}{100} \times (11+1)$

$= 0.75 \times 12$
$= \boxed{9}$

$55 + 9(5)$
$= 55 + 40$
$= \boxed{95}$

$P_{90} = \frac{90}{100} \times (11+1)$

$= 0.90 \times 12$
$= \boxed{10.8}$

$60 + 0.8(60)$
$= 60 + 48$
$= \boxed{108}$

7) 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 200

$P_{20} = 58$
$P_{40} = 71$
$P_{60} = 84$
$P_{80} = 97$
$P_{90} = 170$

(1) $\frac{20}{100} \times (12+1)$

0.2×13
$\boxed{2.6}$

$55 + 0.6(5)$
$55 + 3$
$= \boxed{58}$

(2) $\frac{40}{100} \times (12+1)$

0.4×13
$= \boxed{5.2}$

$70 + 0.2(5)$
$70 + 1$
$= \boxed{71}$

(3) $\frac{60}{100} \times (12+1)$

0.6×13
$= \boxed{7.8}$

$80 + 0.8(5)$
$= 80 + 4$
$\boxed{84}$

(4) $\frac{80}{100} \times (12+1)$

0.8×13
$= \boxed{10.4}$

$95 + 0.4(5)$
$95 + 2$
$= \boxed{97}$

(5) $\frac{90}{100} \times 13$

0.9×13

$= \boxed{11.7}$

$100 + 0.7(100)$

$100 + 70$

$= \boxed{170}$